

# CustomerIQ

## Product Requirements Document (PRD)

### Customer Intelligence & Retention Analytics Engine

Version 1.0 | Current project state: End of Day 1 / Before Day 2

## Document Status

This PRD captures the agreed project scope, completed work, methodological decisions, risks, limitations, and execution roadmap based on the project discussion to date.

## Executive Summary

- CustomerIQ is a portfolio-grade customer intelligence and retention analytics system built on transaction-level retail data. The product transforms raw transactions into customer behavior insights, customer segments, churn-risk signals, retention priorities, recommended marketing actions, and an executive-facing Power BI dashboard.
- The project is deliberately business-first. The intended analytical chain is: Customer Behavior → Insight → Risk → Priority → Action → Monitoring.
- Current project status: the data foundation is complete. The original UCI Online Retail II dataset contained 1,067,371 rows and 8 columns. After documented cleaning, 779,425 transaction rows remain for analysis.

## 1. Product Vision

- Product name: CustomerIQ.
- Working description: AI-Powered Customer Intelligence & Retention Engine.
- Vision: enable a marketer or business decision-maker to understand who their customers are, identify meaningful changes in purchasing behavior, prioritize customers by value and risk, and convert analytics into appropriate actions.
- CustomerIQ is not positioned as a beauty or pure D2C product. The demonstration dataset is a UK online retail dataset with a significant wholesale/B2B component. The framework is therefore positioned as category-agnostic rather than as a fictional beauty brand.

## 2. Problem Statement

- Transaction datasets contain historical purchases but do not directly answer operational questions such as: Which customers are valuable? Which customers have changed their purchasing behavior? Who should be prioritized for intervention? Why is a customer considered at risk? What should marketing or account management do next?
- A model-only solution is insufficient. A churn probability without customer value, explanation, prioritization, and an action framework has limited business usefulness.
- CustomerIQ addresses this by connecting analytical outputs to a decision layer.

## 3. Target Users

- Primary users: marketing analysts, CRM/retention teams, customer analytics teams, e-commerce or retail managers.
- Secondary users: account managers, business analysts, and executives monitoring customer health.
- Primary jobs to be done: understand customer purchasing patterns; identify valuable customers showing signs of disengagement; prioritize limited retention resources; communicate risk in understandable business language; monitor customer health over time.

## 4. Dataset and Evidence Base

- Primary dataset: UCI Online Retail II, used as the demonstration dataset.
- Observed raw dataset results: 1,067,371 rows; 8 columns; date range from 2009-12-01 07:45:00 to 2011-12-09 12:50:00; 5,942 unique customers before cleaning; 53,628 unique invoices before cleaning.
- Columns: Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, Country.
- Data quality findings: 4,382 missing Description values; 243,007 missing Customer ID values, representing 22.77% of the raw dataset; 34,335 exact duplicate rows identified.
- Important framing: the dataset should not be presented as consumer beauty data. Its wholesale/B2B characteristics affect interpretation of AOV, repeat behavior, and purchasing cycles.

## 5. Day 1 Completed Work

- Loaded the raw dataset in Python.
- Validated dataset shape, columns, data types, missing values, duplicate records, date range, customer count, and invoice count.
- Investigated duplicate records before removing them.
- Created a cleaned working dataset by removing exact duplicates, rows without Customer ID, cancelled invoices beginning with C, non-positive quantities, and non-positive prices.
- Converted InvoiceDate to datetime for temporal analysis.
- Created Revenue = Quantity × Price.
- Resulting cleaned dataset: 779,425 rows and 8 original fields before the derived Revenue feature was added.
- Saved the processed dataset for reuse in later notebooks.
- Calculated initial business metrics: Total Revenue £17,374,804.27; Total Customers 5,878; Total Orders 36,969; Average Order Value £469.98.
- Calculated repeat behavior: 4,255 repeat customers, 1,623 one-time customers, and a 72.39% repeat customer rate.

## 6. Interpretation and Caveats

- The £469.98 AOV is a valid calculation from total revenue divided by unique invoices in the cleaned dataset, but it should not be presented as a typical consumer e-commerce benchmark because of the dataset's wholesale/B2B mix.
- The 72.39% repeat customer rate confirms that there is a substantial repeat-purchasing population available for behavioral analysis. It should not be generalized to a typical D2C consumer retention rate.
- Customers with missing Customer IDs were excluded because they cannot be assigned to a customer-level intelligence, segmentation, or churn workflow.

- Cleaning choices must remain documented in the README and notebooks so the portfolio work is reproducible.

## 7. Core Product Workflow

- Stage 1 — Raw transaction ingestion.
- Stage 2 — Validation and cleaning.
- Stage 3 — Customer-order aggregation and purchase behavior analysis.
- Stage 4 — RFM and interpretable customer segmentation.
- Stage 5 — Censoring-aware churn labeling and predictive modeling.
- Stage 6 — Customer value  $\times$  churn risk prioritization.
- Stage 7 — Explainable insights and recommendation logic, with an optional generative AI presentation layer.
- Stage 8 — Power BI dashboard and portfolio presentation.

## 8. Day 2 Objectives: Purchase Behavior

- Create a new notebook for purchase behavior analysis and load only the processed dataset.
- Build a customer-order-level table so that each order can be analyzed as a customer event rather than as a product line.
- Inspect the distribution of order counts before defining frequency tiers.
- Treat any initial tiers such as 2–3, 4–9, and 10+ orders as hypotheses rather than findings. Adjust only after inspecting the distribution.
- Sort each customer's orders chronologically and calculate inter-purchase intervals between consecutive orders.
- Compare interval distributions across data-informed customer frequency tiers.
- Document that customers with fewer orders contribute fewer observed intervals and therefore provide noisier individual estimates of purchasing rhythm.
- End Day 2 with a concrete observation/validation cutoff decision rather than leaving the issue as a concept.

## 9. Right-Censoring and Temporal Methodology

- The dataset ends on 2011-12-09. A customer whose most recent purchase occurs near that end date cannot automatically be treated as churned merely because no subsequent purchase is visible.
- This is a right-censoring problem: the observation window ends before the analyst can observe the customer's full future behavior.
- The project will therefore use a temporal split. Data before a selected cutoff will form the observation window used for features and behavior analysis. Data after the cutoff will serve as a validation/look-forward window used to observe whether eligible customers return.
- The cutoff will be selected after inspecting the actual inter-purchase interval distribution. The validation period must be long enough to capture a meaningful portion of reasonable return intervals.
- The exact cutoff date is not yet locked. Selecting and documenting it is a Day 2 deliverable.

## 10. Churn Definition Strategy

- Churn will not initially be hardcoded as 90 days of inactivity.

- A defensible threshold will be informed by observed customer purchase rhythms while accounting for customer heterogeneity and the finite end of the dataset.
- The project should avoid using future information when constructing features for prediction.
- One-time buyers may be excluded from the repeat-customer churn modeling population because a customer with only one observed purchase has not demonstrated an established repeat-purchase relationship. This rule will be validated and documented when the final modeling cohort is created.
- The final churn definition must be simple enough to explain in an interview and reproducible from the data.

## 11. Customer Segmentation Plan

- RFM analysis will calculate Recency, Frequency, and Monetary value using an explicit analysis date.
- Segments will be interpretable and business-oriented rather than relying solely on opaque clustering.
- Potential segment labels include Loyal/High Value, New or Emerging, Discount/Price Sensitive if valid features become available, At-Risk, and Inactive. Segment definitions will be based on available evidence rather than forced labels.
- K-Means or other clustering is optional, not a core requirement. The MVP prioritizes interpretable RFM-based segmentation.

## 12. Churn Modeling Plan

- Start with an interpretable baseline such as Logistic Regression.
- Compare against a stronger non-linear benchmark such as Random Forest if it provides useful incremental value.
- Use a time-aware train/validation strategy where possible to reduce leakage.
- Evaluate more than accuracy. Candidate metrics include ROC-AUC, precision, recall, PR-AUC where class imbalance warrants it, confusion matrix, and business-focused performance on high-value customers.
- A useful business metric is recall among high-value customers: the system should avoid missing valuable customers who are actually at risk.
- Model output will be probability-based and later combined with customer value rather than treated as a standalone decision.

## 13. Retention Priority Framework

- Core concept: Churn risk alone is not enough. A low-value customer with high risk should not automatically receive the same intervention as a high-value customer with high risk.
- CustomerIQ will combine customer value signals with predicted churn risk to create practical priority categories.
- Illustrative logic: High Value + High Risk → priority retention; High Value + Low Risk → loyalty/cross-sell; Low Value + High Risk → low-cost reactivation; Low Value + Low Risk → monitor or lower-priority nurture.
- The exact scoring formula will be chosen after RFM and churn outputs are available.

## 14. Recommendation and GenAI Layer

- The recommendation engine should begin with deterministic business rules so recommendations are traceable and testable.
- A generative AI layer may then convert model drivers and customer context into concise explanations and marketer-friendly recommended actions.

- Example output style: explain the main behavioral signals associated with risk and recommend an intervention consistent with customer value and economics.
- The GenAI layer must not be represented as evidence that an intervention will reduce churn. The dataset does not contain campaign treatment/outcome data sufficient to prove causal impact.

## 15. Dashboard Requirements

- Tool: Power BI.
- Executive KPI view should remain focused. Candidate MVP KPIs: total active customers, at-risk customers, high-value at-risk customers, observed churn/inactivity rate under the final definition, model quality such as ROC-AUC, and high-value customer recall.
- Additional views: customer segments, churn-risk distribution, top model drivers, priority customers, recommended action categories, and trend analysis where the temporal methodology supports it.
- The dashboard should answer business questions, not simply display every available chart.

## 16. Technology and Repository Structure

- Primary stack: Python, pandas, SQL, Power BI, Git/GitHub, and an optional generative AI layer.
- Current project structure concept: CustomerIQ/Data/Raw, CustomerIQ/Data/Processed, notebooks, sql, dashboard, screenshots, README.
- Notebooks should be numbered by analytical stage, for example 01\_data\_validation.ipynb and 02\_purchase\_behavior.ipynb.
- Every major stage should produce a reusable artifact and a documented conclusion.

## 17. SQL Scope

- SQL will be used to demonstrate customer and transaction analysis rather than duplicated purely for the sake of having SQL in the project.
- Potential analyses: customer order counts, revenue aggregation, repeat customer identification, recency/frequency calculations, high-value customer ranking, and segment summaries.
- SQL outputs should align with the Python and Power BI business logic.

## 18. Success Criteria

- MVP success means: reproducible data pipeline; documented data-quality decisions; defensible temporal methodology; interpretable customer segmentation; leakage-aware churn model; value × risk prioritization; explainable recommendations; focused Power BI dashboard; clear GitHub README and portfolio narrative.
- Strong project success additionally means: transparent limitations, model comparison, business-focused evaluation, screenshots/demo, and the ability to explain every major design choice in an interview.

## 19. Known Risks and Mitigations

- Risk: right-censoring near the dataset end. Mitigation: observation/validation temporal split and explicit cutoff.
- Risk: one global churn threshold misrepresents heterogeneous customers. Mitigation: inspect intervals by frequency tier and document the chosen simplification.
- Risk: B2B/wholesale behavior is presented as consumer behavior. Mitigation: honest dataset framing and careful interpretation of AOV and repeat rate.

- Risk: recommendation effectiveness cannot be proven. Mitigation: state clearly that recommendations are decision rules/hypotheses, not causal results.
- Risk: scope creep. Mitigation: Logistic Regression + Random Forest + RFM + decision layer as the core; no mandatory deep learning, XGBoost, survival analysis, K-Means, or complex agent architecture.
- Risk: low-order customers have noisy interval estimates. Mitigation: acknowledge interval-count imbalance and summarize behavior at appropriate cohort/tier levels.
- Risk: synthetic CRM features could be mistaken for real observations. Mitigation: if added later, label them explicitly as synthetic augmentation and separate them from original fields.

## 20. Project Roadmap

- Completed — Foundation: project framing, dataset selection, validation, cleaning, processed dataset, revenue feature, initial KPIs, repeat-customer analysis.
- Current — Day 2: customer-order table, order-count distribution, data-informed tiers, inter-purchase intervals, right-censoring-aware cutoff decision.
- Next — Churn methodology: finalize modeling cohort, observation features, validation labels, and leakage checks.
- Then — RFM and segmentation.
- Then — churn modeling and evaluation.
- Then — value  $\times$  risk prioritization and recommendation framework.
- Then — GenAI explanation layer.
- Then — Power BI dashboard.
- Final — GitHub README, case study, screenshots, demo, and interview preparation.

## 21. Current Progress Assessment

- Foundation phase: effectively complete for the initial scope.
- Overall product completion: approximately 12–15% at the current point.
- The project has not yet built the visible intelligence, prediction, prioritization, recommendation, or dashboard layers. The completed work is nevertheless substantive because it validates the data and prevents downstream methodology from being built on an unexamined dataset.

## 22. Interview Narrative

- Short version: I built CustomerIQ, a customer intelligence and retention analytics system. I started with raw transaction data, validated and cleaned it, then analyzed customer purchase behavior to define a defensible, time-aware churn methodology. From there I use RFM and interpretable segmentation, predict customer risk, combine risk with customer value to prioritize interventions, and surface explanations and recommended actions in a Power BI dashboard.
- Important limitation to state: the project demonstrates prioritization and recommended interventions; it does not claim causal proof that those interventions reduce churn because no treatment/outcome campaign data is available.

## 23. Immediate Next Action

- Do not redesign the project again before execution.

- Start Day 2 with a single small action: create 02\_purchase\_behavior.ipynb and load the processed dataset.
- Then proceed sequentially: customer-order table → order-count distribution → tier decision → inter-purchase intervals → interval comparison → concrete temporal cutoff → methodology notes.
- Day 2 is complete only when a documented cutoff decision and its rationale exist.